Huawei Voice assistant
Huawei mobile cheif Richard Yu says the company will have its own voice assistant soon in China. Learn more here: **http://dgit.in/HuaweiAI**

Video calls for Signal
Currently available in the open beta with the new update, you can now make encrypted video calls using Signal. **http://dgit.in/encryptedV**

HOW WE TESTED

In testing smartphones, benchmarks are the first things to consider. They do not always represent real world performance, but they provide a good way to objectively compare the performance of smartphones. Our benchmarks of choice, that you'll see in upcoming pages, are AnTuTu, GFXBench Car Chase, 3D Mark Slingshot, 3D Mark Unlimited and Geekbench 4 Single and Multi Core. However, when reviewing phones, we do use other tools and benchmarks to better gauge performance. All of these readings and observations figure into the final score for the product.

Further, heat dissipation is often a major issue amongst cheaper smartphones. In our tests, we tested each smartphone for the amount of heat they generated. All the tests were performed in current room temperatures of about 27 degrees Celsius, and both gaming and video recording were measured. We used Asphalt 8 and Injustice: Gods Among Us, two resource intensive games to measure the temperature.

When measuring temperature, we played the same games for 15 minutes on each device, using a heat gun to measure temperature on the device's body. Internal temperatures were also recorded. Similarly, video recording tests were conducted over 10 minutes.

It is worth mentioning that we split the whole test into three parts: Features, Performance and Design. The Performance section includes not only the raw power a phone has, but also its camera quality, battery life and call quality. The features part is self explanatory and the design section takes into consideration the display quality, materials used and even the user interface of the device.

To establish a common ground, all phones had exactly the same apps installed on them. This means whatever extra load is added on the device, would come from its own system and user interface or pre-installed apps.

Next, camera quality was measured in different light conditions. We keep a separate score for low light photography, since companies tend to market that aspect of devices often. However, all other conditions, including bright sunlight, indoors and others, are part of the normal light score that we arrive at. All phones were tested using the same images, taken from each device. It is also worth noting that the front camera hasn't been considered, which is also why we left phones like the Vivo V5 and Oppo F1s out of this comparison. We are not prone to taking selfies ourselves, but if you do want one such comparison, do write to us, we are indeed equipped to present an objective comparison of front cameras.

Speed, battery life and camera quality aside, we did consider aspects like build quality, touch performance, the UI design, weights and ergonomics in general. Thinness of the phones was taken into account, as was the ease of use of its user interface.

Lastly, the pricing of each of the device has also been accounted for. While the best performer is chosen irrespective of price points, the best buy award is directly related to the price. We measure the overall performance per rupee, that you can get out of a device, with algorithms that are so complex that anyone more intelligent than a monkey can understand them.

So, without further ado, here's all you need to know about smartphones between ₹10,000 and ₹20,000.

raw power and speed. Its benchmark scores are nearly double, and sometimes triple, of what other smartphones can achieve. It is fast, responsive and snappy, like few smartphones can be. The Snapdragon 820 churns out oodles of raw power, leaving all of its competition behind.

However, while that does make the Lenovo Z2 Plus our best performer, we would note that in terms of real world performance, the Coolpad Cool 1 is just as great. The Lenovo Z2 Plus is faster, for the sake of comparison, but in real world terms, 4GB RAM and the Snapdragon 652 are also very hard to argue against.

What the Lenovo Z2 Plus and Coolpad Cool 1 promise is a lag free experience. While the former showed practically no noticeable lags, the latter was ever so slightly stuttery. What Lenovo has ideally done, is produce a smartphone that is just impossible to argue against, for just one spec. However, the Z2 Plus does belong to the Zuk series, and there's an enthusiast approach here.

In doing so, you can expected some UI issues. While the unit used for this comparison was sent to us by Lenovo, we considered another store-bought unit of the Lenovo Z2 Plus as well, and there are certain elements about that device that regular users may not enjoy. That, however, is a trade-off on phones that are meant for an enthusiast community, where the OS is supposed to be ever evolving and feature rich.

In terms of performance, the Xiaomi Redmi Note 4 seems to be the most dependable device, although we have seen MiUI updates cause issues on Xiaomi phones before. The Lenovo P2 also has a very good overall experience, thanks to the 14nm Snapdragon 625 SoC. The phone is of course battery centric, but few would be able to question its smoothness.

Camera

It's a wonderful world we live in nowadays. You can buy a dual-camera smartphone at well below ₹20,000. If that isn't improvement then we do not know what is. Amongst the many "improvements" that smartphone makers have tried to bring about in cameras, dual-cameras are perhaps the first to show real impact.

You would expect the Honor 6x or the Coolpad Cool 1 to win the prize here, right? Well, think again. Our tests suggest the little known ZTE Nubia Z11 Mini is the best phone for those who want a camera over everything else. ZTE corrected the shutter lag issue from the Z9 Mini, producing a phone that can take very good images in all light conditions, considering its price.

The Nubia Z11 Mini doesn't mess up the contrast ratio and produces well balanced colours, something that many smartphones today are unable to do. In low light, the Nubia Z11 Mini's camera software tries to brighten photos artificially, making colours a tad bleak in the process. However, the overall results are still far better than its